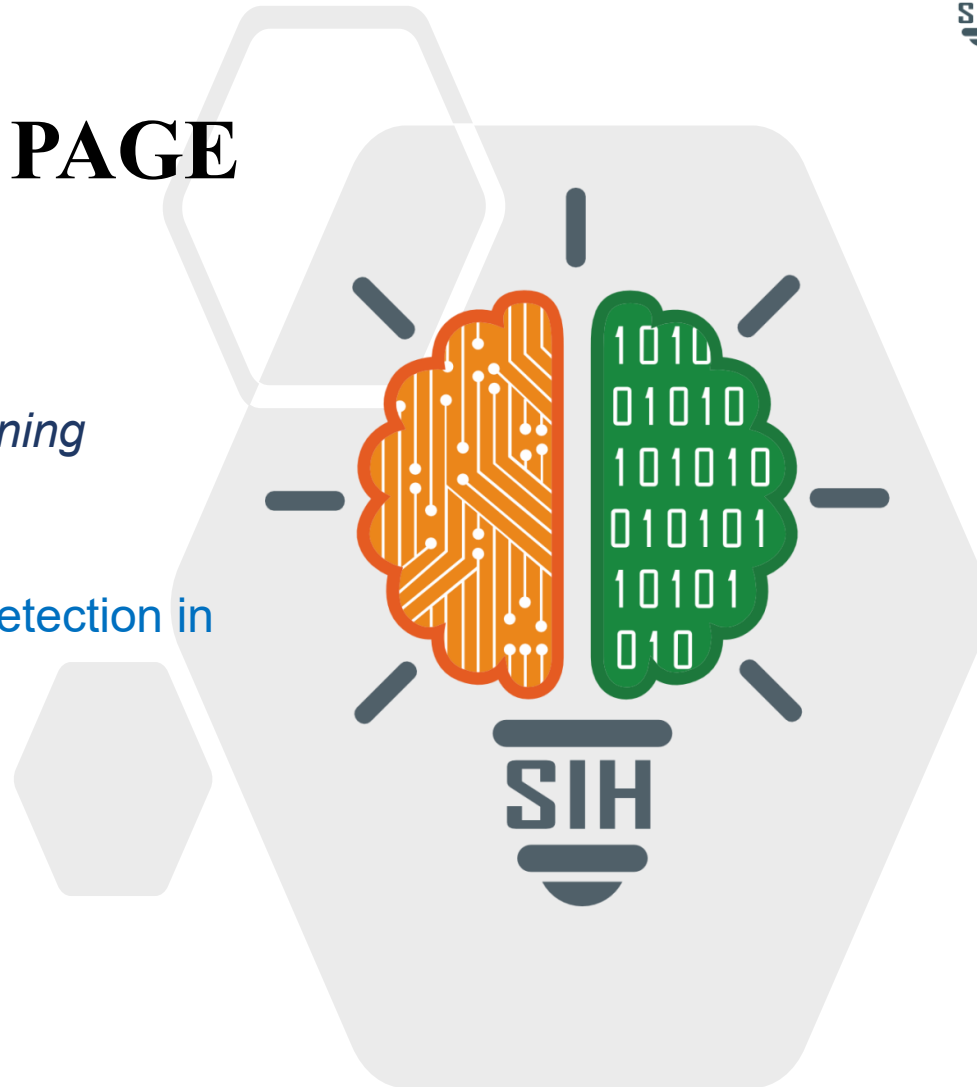


TITLE PAGE

AgniDrishti

Static screening → trajectory-aware burn-in screening

- **Problem Statement ID** – 26170
- **Problem Statement Title**- AI-Driven Anomaly Detection in Component Burn-In & Screening
- **Theme**- Smart Automation
- **PS Category**- Software
- **Team ID**- [TEAM ID]
- **Team Name**- Regnum Carya

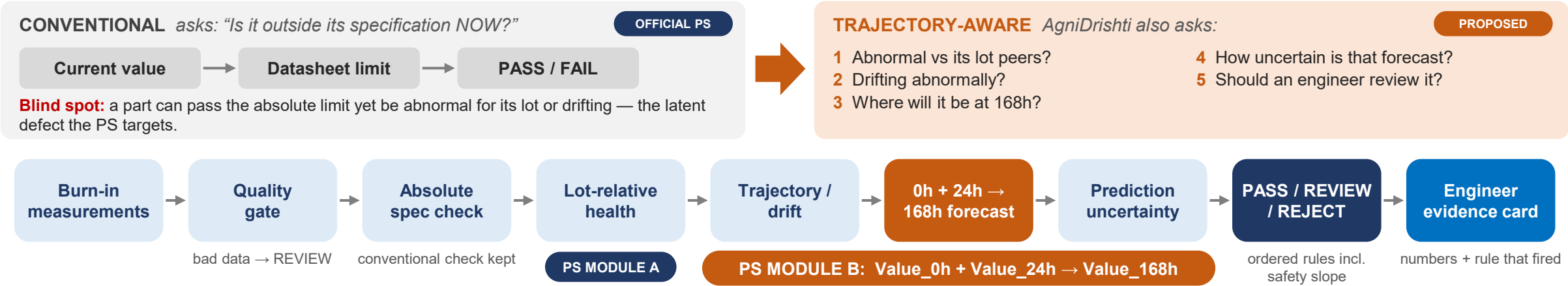


Static screening → trajectory-aware burn-in screening

Proposed Solution (Describe your Idea/Solution/Prototype)

Labels: OFFICIAL PS PROPOSED CONCEPTUAL

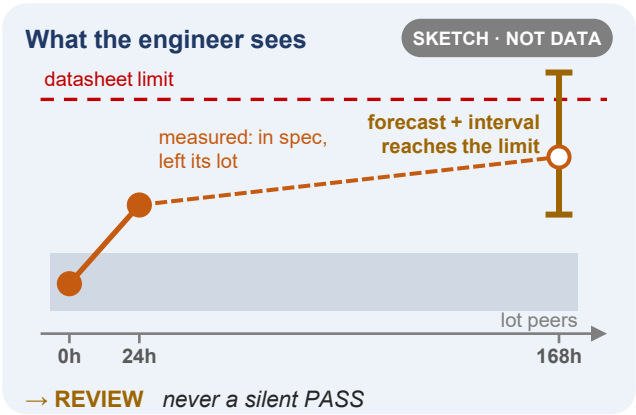
Detailed explanation of the proposed solution



How it addresses the problem

PS: Module A → lot health · Module B → forecast · doubt → REVIEW · explain → card

ENGINEER EVIDENCE CARD <i>conceptual mock-up · no real data</i>	
Component <component_id> · Lot <lot_id> · checkpoint 24h	
Current specification	PASS inside datasheet limit
Lot behaviour	ABNORMAL far from lot median
Trajectory (0→24h)	DEGRADING drift unusual for its lot
Predicted Value_168h	<point forecast>
Prediction uncertainty	<prediction interval>
Decision	REVIEW
Reason	rule fired: forecast interval reaches limit; lot outlier



Innovation and uniqueness of the solution

“Combines static specification screening, lot-relative behaviour, early trajectory analysis, 168h forecasting, uncertainty and explainable review in one screening workflow.”

Transparent by design: the explanation is the decision evidence itself.

Individual techniques are established; the differentiation is their PS-specific integration and operational workflow.

Research foundations: lot-relative limits (AEC-Q001) and burn-in drift limits (ESCC 9000) informed the design — no compliance claimed.

Technologies to be used (e.g. programming languages, frameworks, hardware)

Python 3.11+

pandas

NumPy

pytest

robust stats: median · IQR · MAD

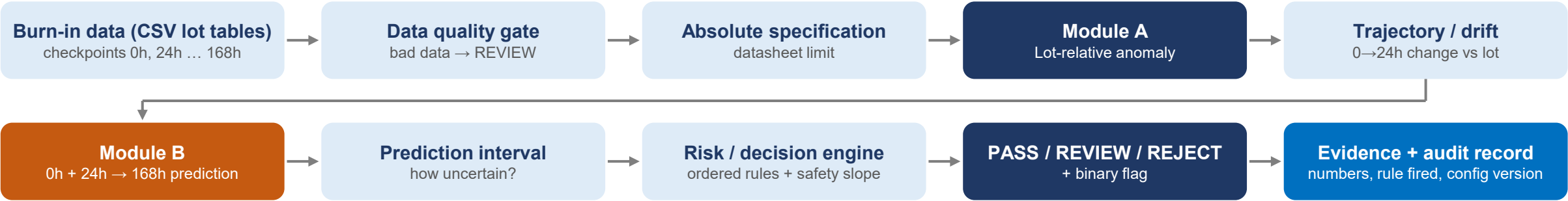
least-squares regression

versioned YAML config

offline · CPU-first

In use now: Python, pandas, NumPy, pytest — data schema + quality gate, unit-tested. **ML comparators (scikit-learn):** Isolation Forest, gradient-boosted trees — adopted where they beat the baseline.

Methodology and process for implementation (Flow Charts/Images/ working prototype)



1

Lot anomaly detection

Robust (median-based) distance of each part's level and 0→24h drift from its own lot; Isolation Forest as ML comparator.

ESTABLISHED METHOD

2

ML forecasting ladder

Persistence → linear → regression on (v0, v24) → gradient-boosted trees. A model stays only if it lowers MAE on held-out lots.

PROPOSED

3

Uncertainty

A range around every 168h forecast (prediction interval), calibrated on held-out lots — not a single overconfident number.

ESTABLISHED METHOD

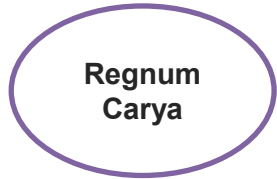
4

Explainability

Every decision carries its numbers and the ordered rule that fired (incl. the safety-slope rule). No black-box score.

PROPOSED

Validation built in: lot-grouped splits · frozen final test lots · automated leakage tests (no 96h/168h value used at 24h) · ablation: each layer stays only if removing it measurably hurts.
All formulas (lot score, drift rate, safety slope) are proposed engineering definitions where the PS is silent — documented in our design records.



FEASIBILITY AND VIABILITY



Potential challenges and risks	Strategies for overcoming these challenges
Official dataset unavailable	Controlled synthetic PS-shaped data (always labelled) + external public aging data — methodology validation only + official-data-ready pipeline
Only Value_0h and Value_24h at decision time	Simple, stable predictors + prediction interval + REVIEW path
Lot contamination / mixed populations	Data-quality checks + robust (median-based) statistics
Whole lot may degrade together	Absolute safety backstop in addition to lot-relative analysis
Safety slope / Anomaly Score not defined in the PS	Configurable engineering definition + sensitivity analysis + organiser clarification
Data leakage / optimistic results	Lot-grouped splits + frozen test lots + no future features (automated tests)
Deployment in controlled facilities	Offline, CPU-first architecture; measurement data stays on site

Analysis of the feasibility of the idea

Technical

Robust statistics + least-squares regression on small lot tables; no GPU.

Operational

Reads checkpoint measurements, returns PASS / REVIEW / REJECT; engineer can override.

Financial

Open-source stack on a standard CPU workstation; no licence or cloud cost.

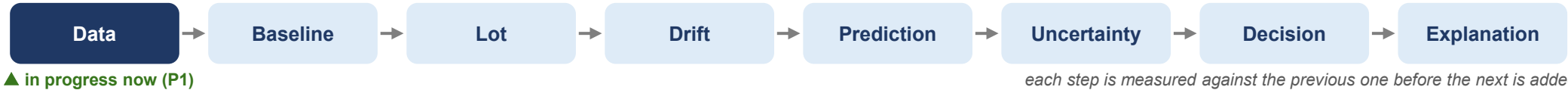
Auditable

Every threshold in versioned config; each decision reproducible from its record.

Already started

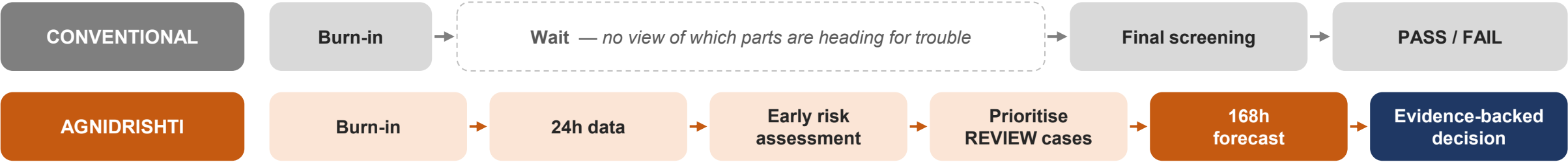
Canonical data schema + data-quality gate implemented and unit-tested.

Incremental build path



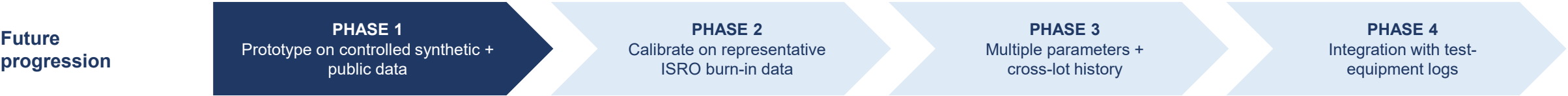
Potential impact on the target audience

For: component screening / QA and reliability engineers, space-grade electronics



Adds an early, evidence-backed view at 24h; the engineer stays in the loop for every REVIEW and REJECT.

Benefits of the solution (social, economic, environmental, etc.)



Quantitative gains will be measured after representative-data validation. No results or impact figures are claimed at the idea stage.

Details / Links of the reference and research work

OFFICIAL

Official

SIH 2026 PS 26170 — AI-Driven Anomaly Detection in Component Burn-In & Screening (ISRO). sih.gov.in/sih2026PS

SIH 2026 Guidelines; SIH 2026 Idea Presentation Format — sih.gov.in/letters/2026/

RELIABILITY FOUNDATIONS

Research · no compliance claimed

AEC-Q001 Rev-D, Guidelines for Part Average Testing, Automotive Electronics Council, 2011

ESCC Generic Specification No. 9000, Issue 10, ESA, 2018

ESCC Detail Specification No. 9202/045, Issue 6, ESA, 2020

DATA

External public dataset · not SIH/ISRO data

NASA PCoE Data Repository — MOSFET Thermal Overstress Aging (Celaya, Saxena, Saha, Goebel)

NASA PCoE — IGBT Accelerated Aging (Celaya, Wysocki, Goebel, 2009)

Used for methodology validation only.

PRIOR ART

Research

Burn-in ML: I. Ahmed, P. Baraldi, E. Zio, H. Lewitschnig, Comput. Ind. Eng. 204, 111115, 2025 · doi:10.1016/j.cie.2025.111115

Lot health for burn-in: L. Langenberg, R. Pätzold, A. Khalid, ESREL 2023

Burn-in outliers: US 8,010,310 B2, Method and apparatus for identifying outliers following burn-in testing, 2011

METHOD

Research

Conformal prediction: J. Lei et al., JASA 113(523), 2018 · doi:10.1080/01621459.2017.1307116

Grouped (lot) data: R. Dunn, L. Wasserman, A. Ramdas, JASA 118(544), 2023 · doi:10.1080/01621459.2022.2060112

Degradation modelling: C. J. Lu, W. O. Meeker, Technometrics 35(2), 1993 · doi:10.1080/00401706.1993.10485038

Reject option (REVIEW): C. K. Chow, IEEE Trans. Inf. Theory 16(1), 1970 · doi:10.1109/TIT.1970.1054406

How to read these: Official sources define the problem. Reliability and prior-art sources informed the design choices. Method sources underpin the proposed statistics. NASA data is an external public dataset for methodology checks — it is not SIH or ISRO data. No official PS dataset has been published.